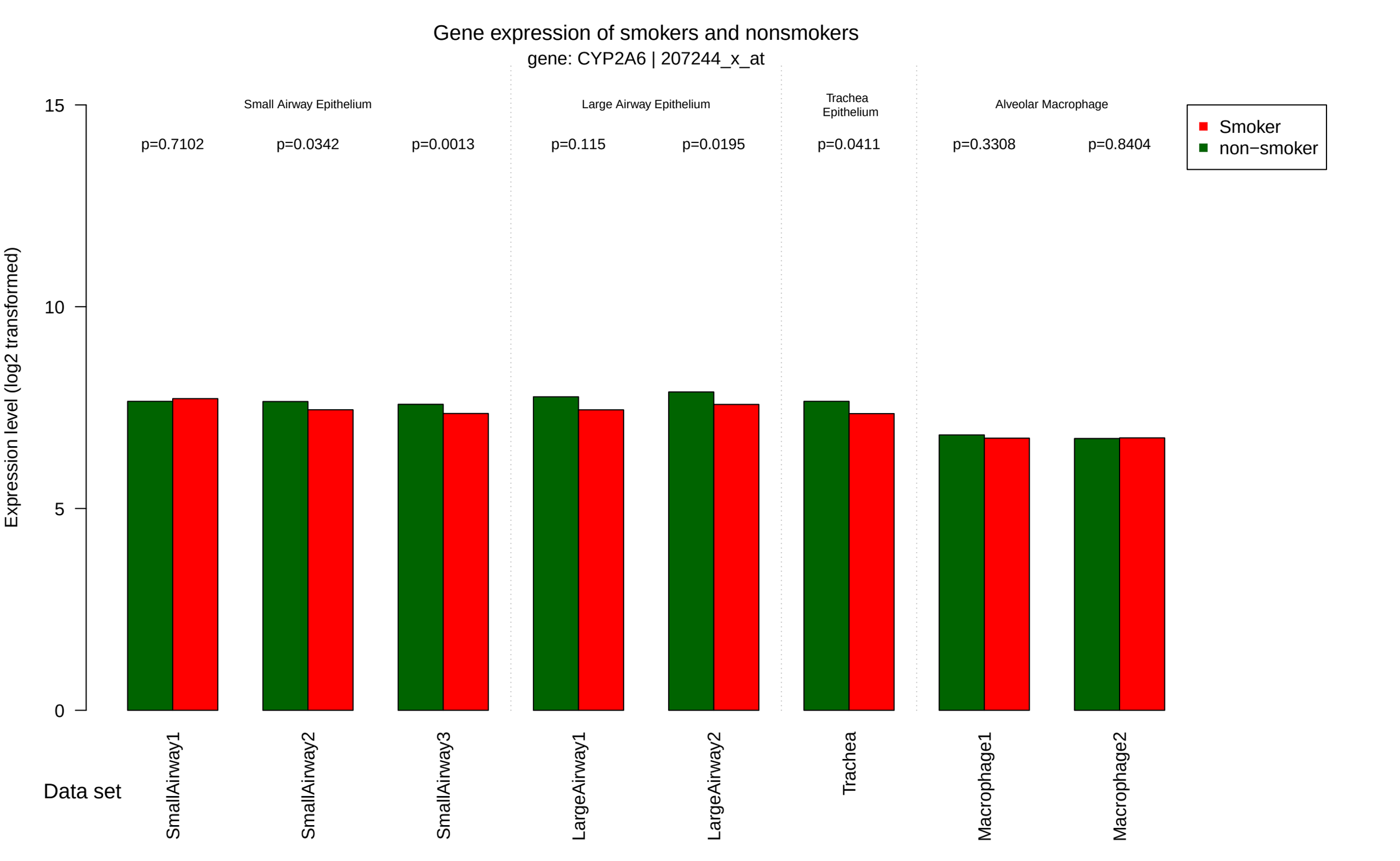
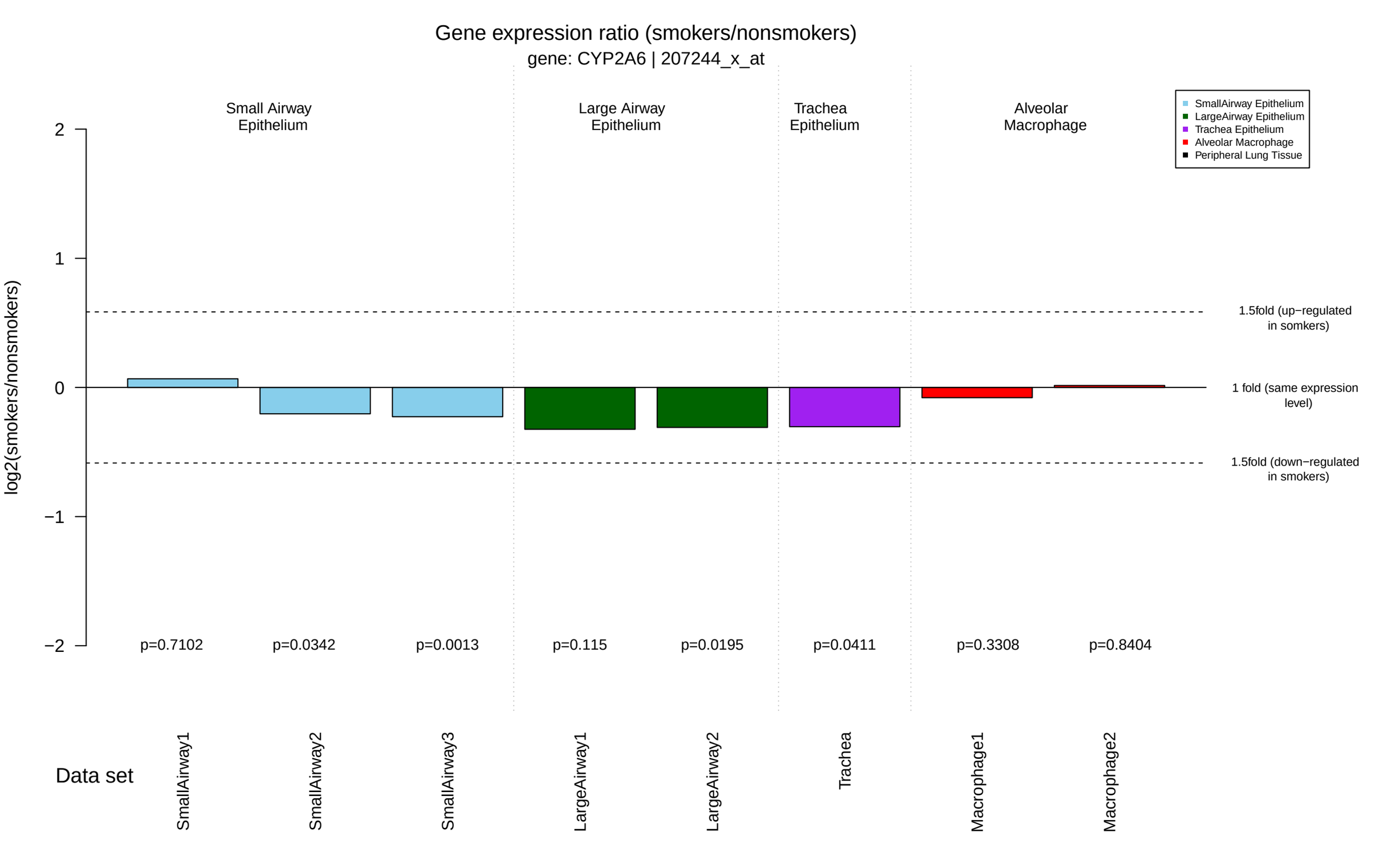
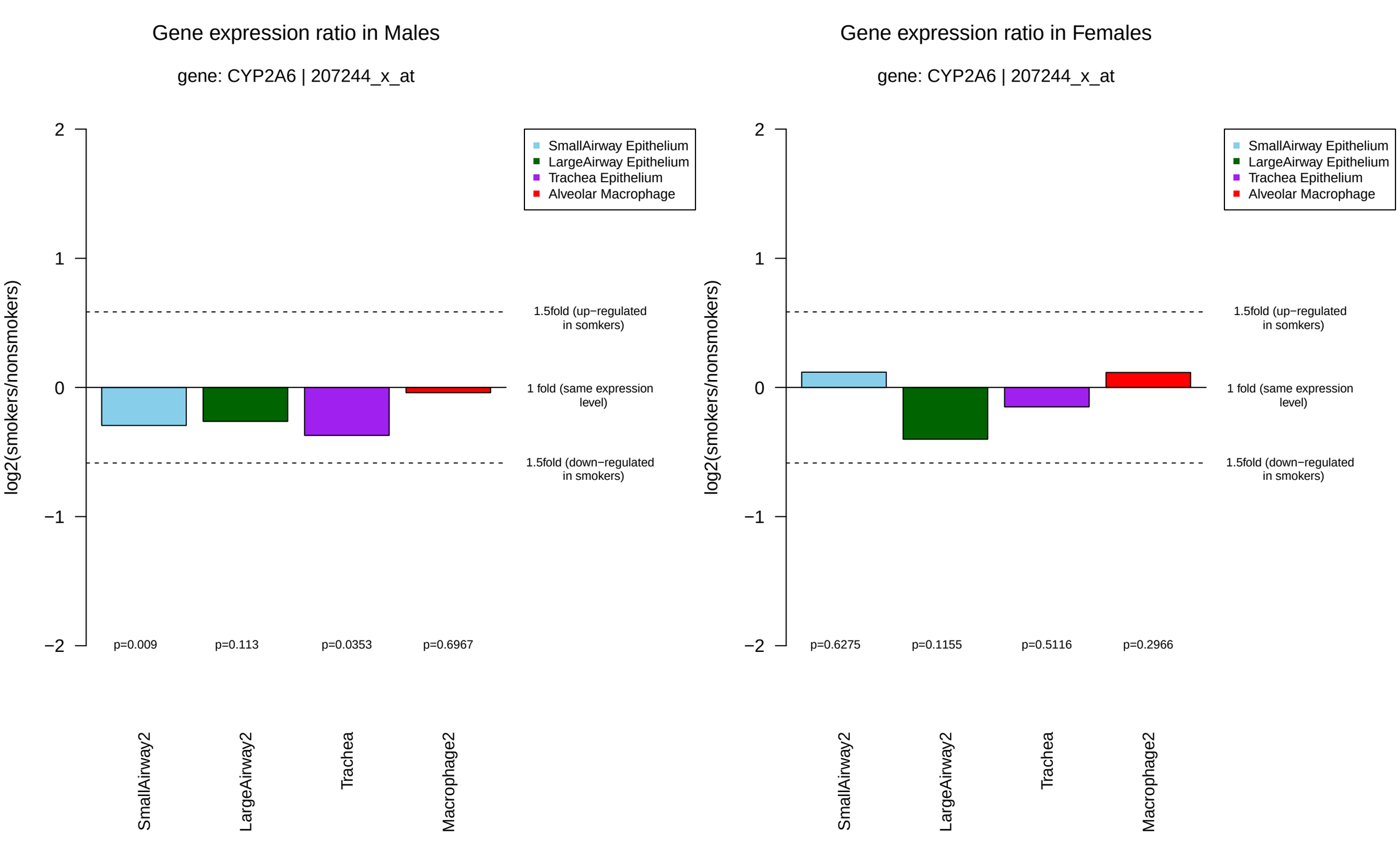
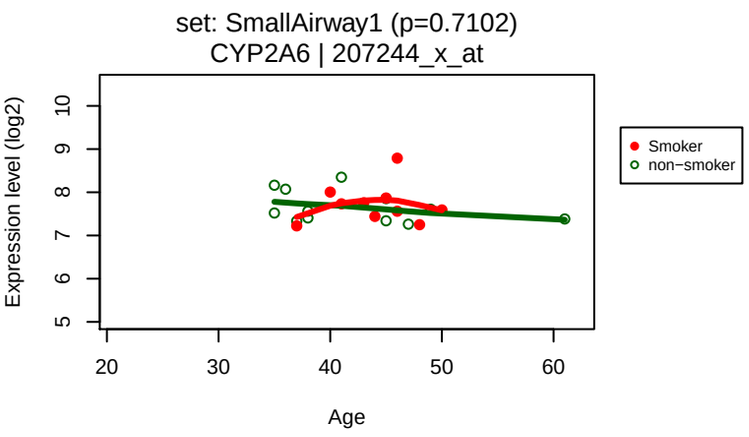
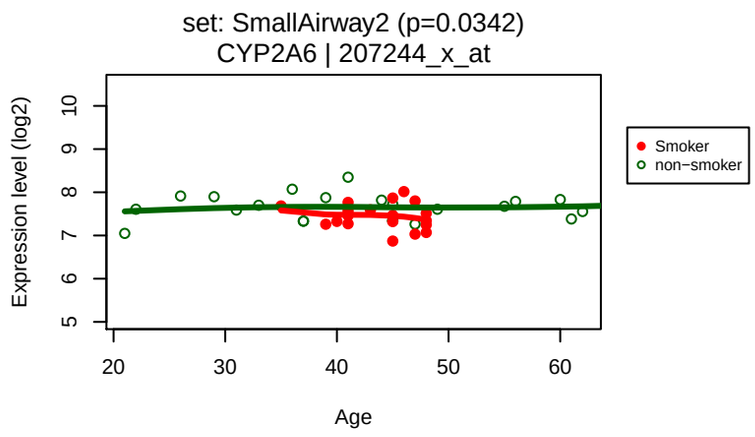
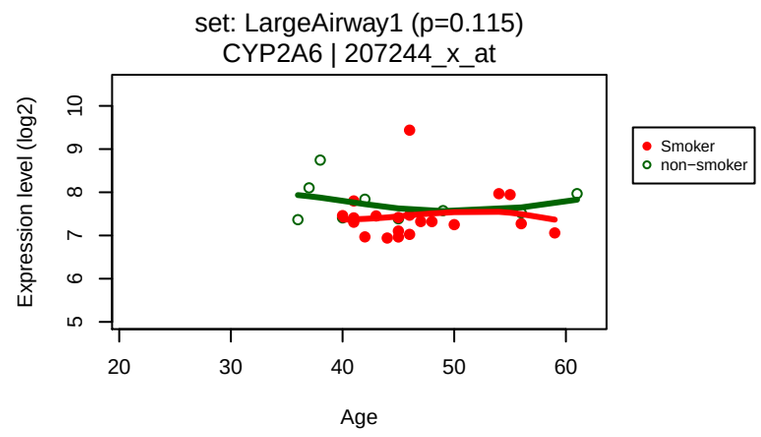

Gene expression of smokers and nonsmokers
gene: CYP2A6 | 207244_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7102
p=0.0342
p=0.0013
p=0.115
p=0.0195
p=0.0411
p=0.3308
p=0.8404
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

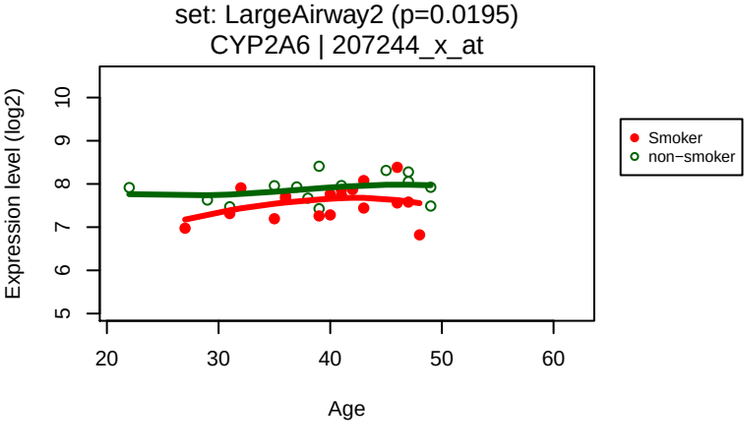

Gene expression ratio (smokers/nonsmokers)
gene: CYP2A6 | 207244_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7102
p=0.0342
p=0.0013
p=0.115
p=0.0195
p=0.0411
p=0.3308
p=0.8404
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

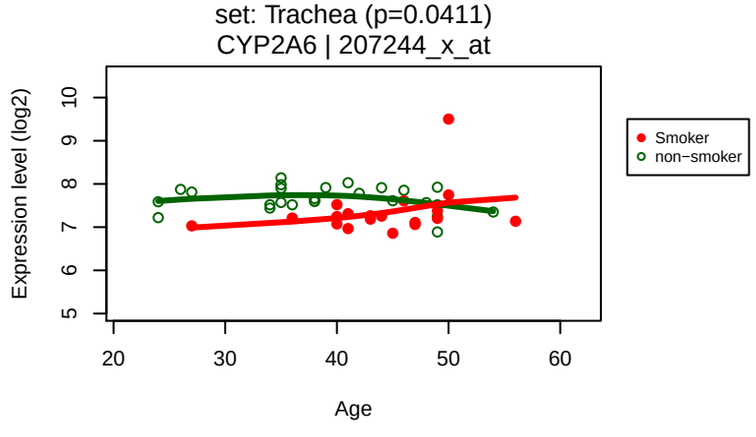

Gene expression ratio in Males
gene: CYP2A6 | 207244_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.009
p=0.113
p=0.0353
p=0.6967
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2A6 | 207244_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6275
p=0.1155
p=0.5116
p=0.2966
SmallAirway2
LargeAirway2
Trachea
Macrophage2

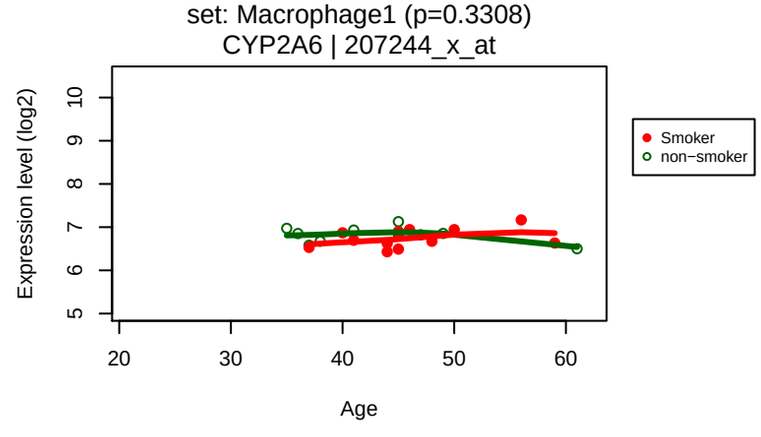

set: SmallAirway1 (p=0.7102)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker

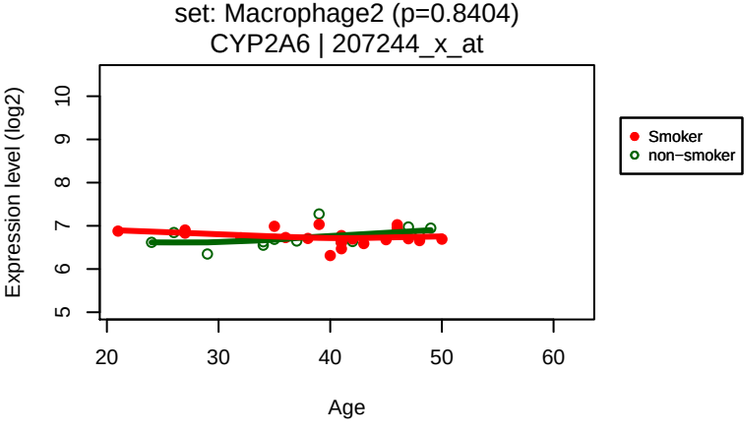

set: SmallAirway2 (p=0.0342)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker

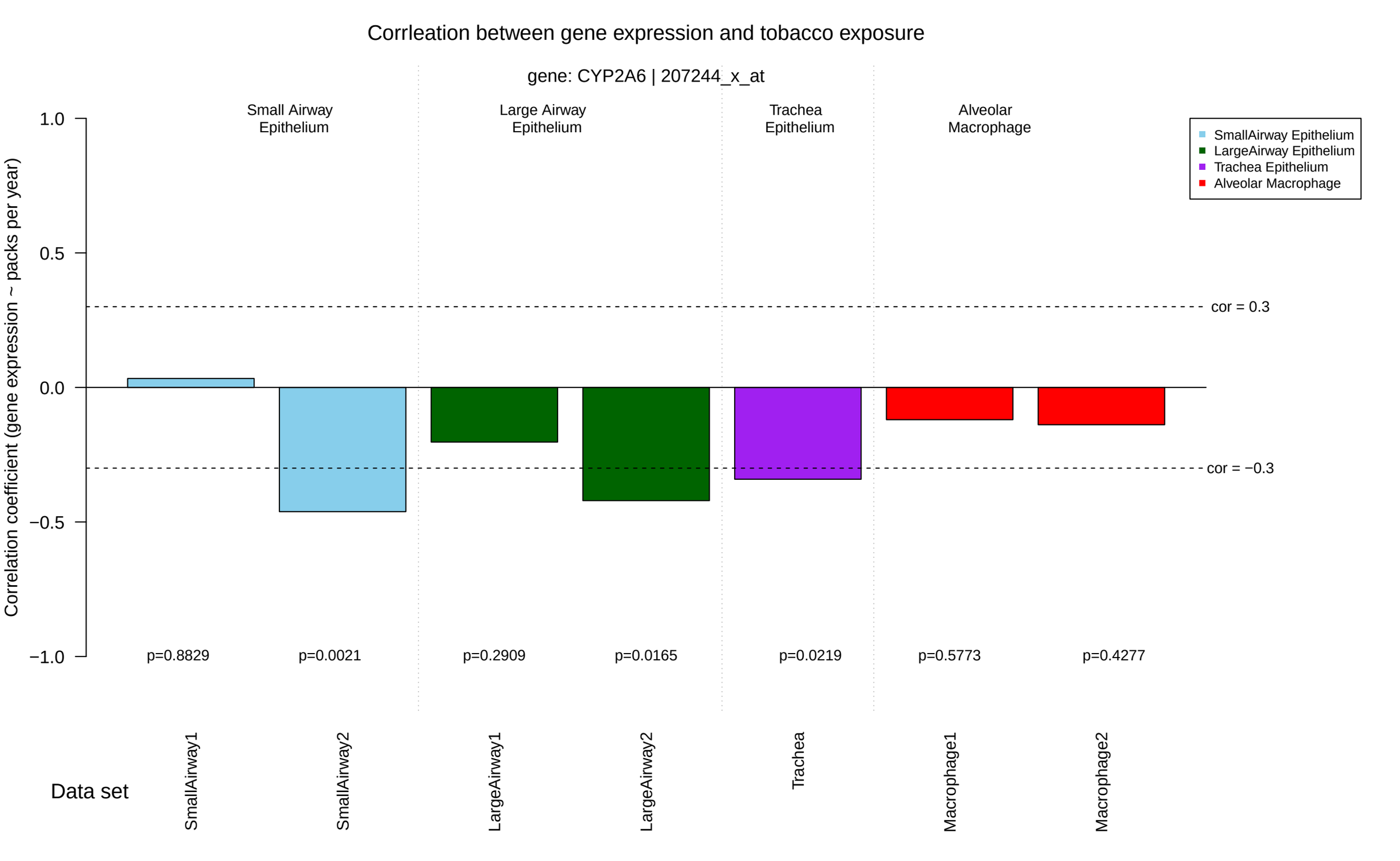

set: LargeAirway1 (p=0.115)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0195)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0411)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3308)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8404)
CYP2A6 | 207244_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2A6 | 207244_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8829
p=0.0021
p=0.2909
p=0.0165
p=0.0219
p=0.5773
p=0.4277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2